

Comparative Study of Documentation Systems for CMS with an AI-Assisted Docs-as-Code Prototype

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KAYST Academy Internshio at CERN . CMS Collaboration . Summer 2026

ABSTRACT

The CMS experiment at CERN has relied on TWiki for over two decades to manage approximately 44,677 documentation pages. This project modernises the documentation ecosystem through two contributions: a custom TWiki-to-GFM converter, twiki2gfm –achieving 98.6% conversion success in 24 minutes, and an AI assisted RAG pipeline combining hybrid retrieval and LLM-based generation. The prototype was deployed on the CERN network to support semantic search and documentation-grounded answers.

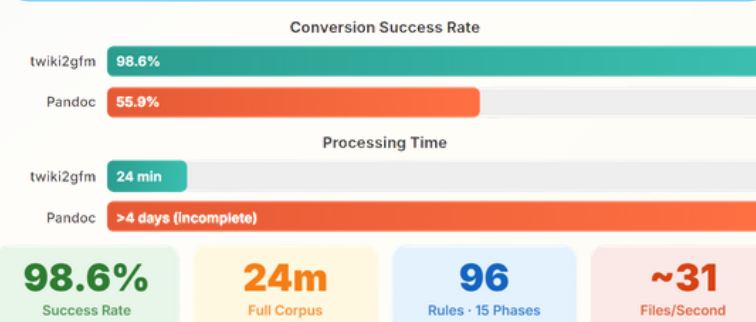
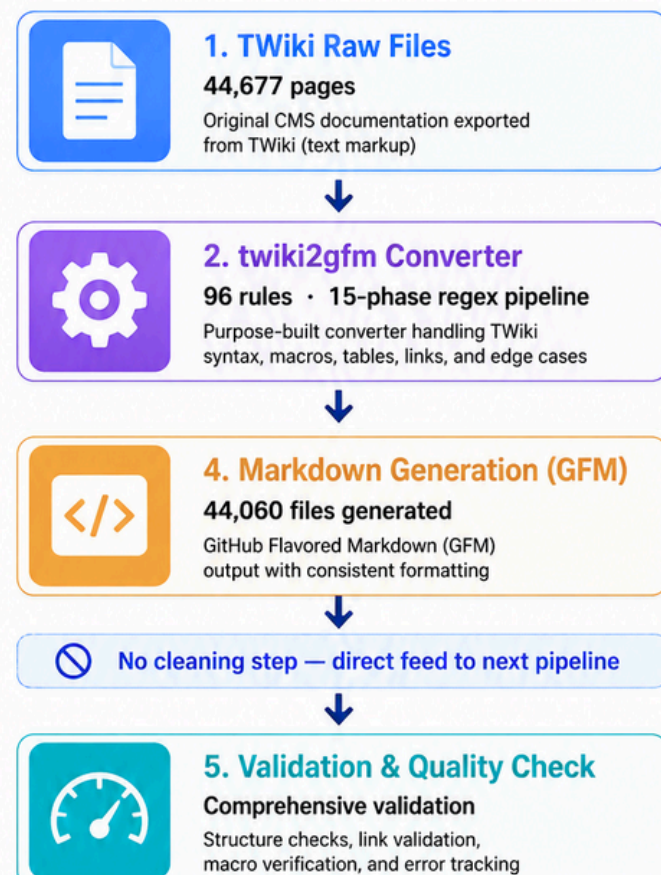
Objectives

- Modernise legacy CMS documentation**
Convert TWiki content into clean, portable GFM
- Build reliable documentation tooling**
Develop and evaluate a custom converter at scale.
- Enable semantic discovery**
Use AI-powered RAG to retrieve relevant CMS knowledge.
- Deliver a practical AI assistant**
Provide grounded documentation answers through a CERN deployed interface

METHODOLOGY

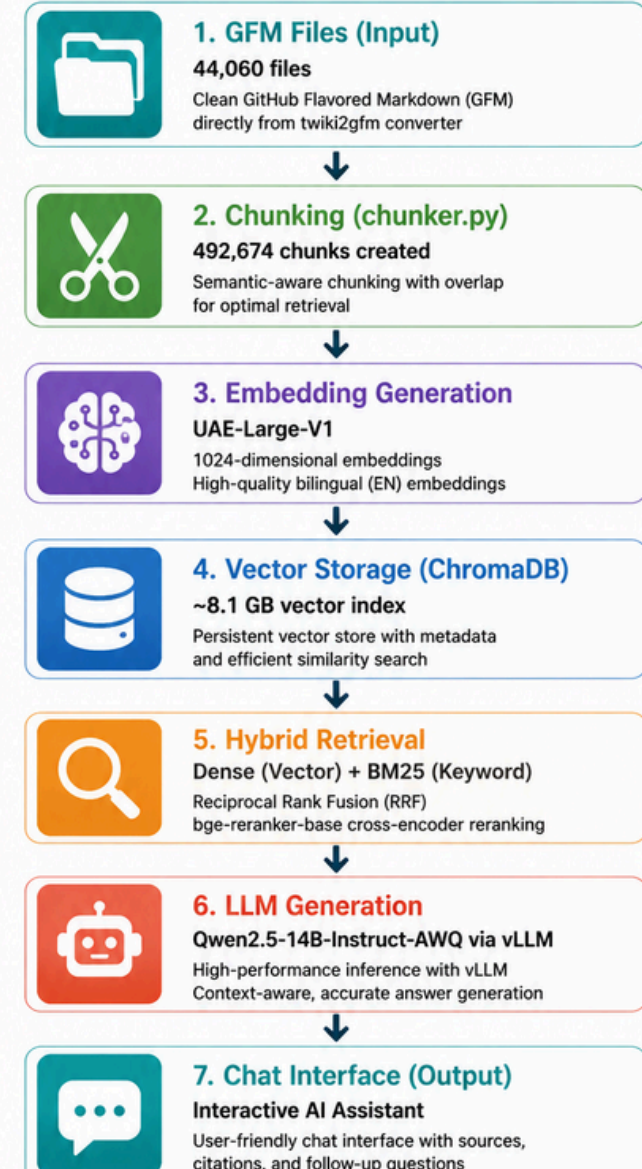
PART 1 : TWIKI TO GFM CONVERSION

High accuracy, High Speed Conversion



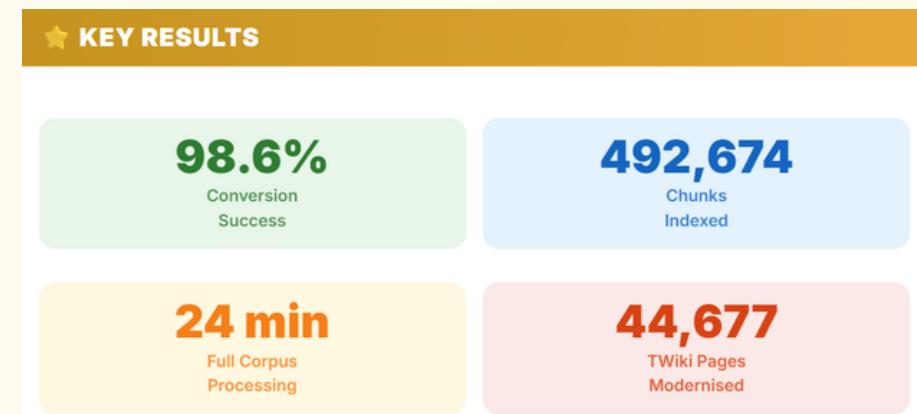
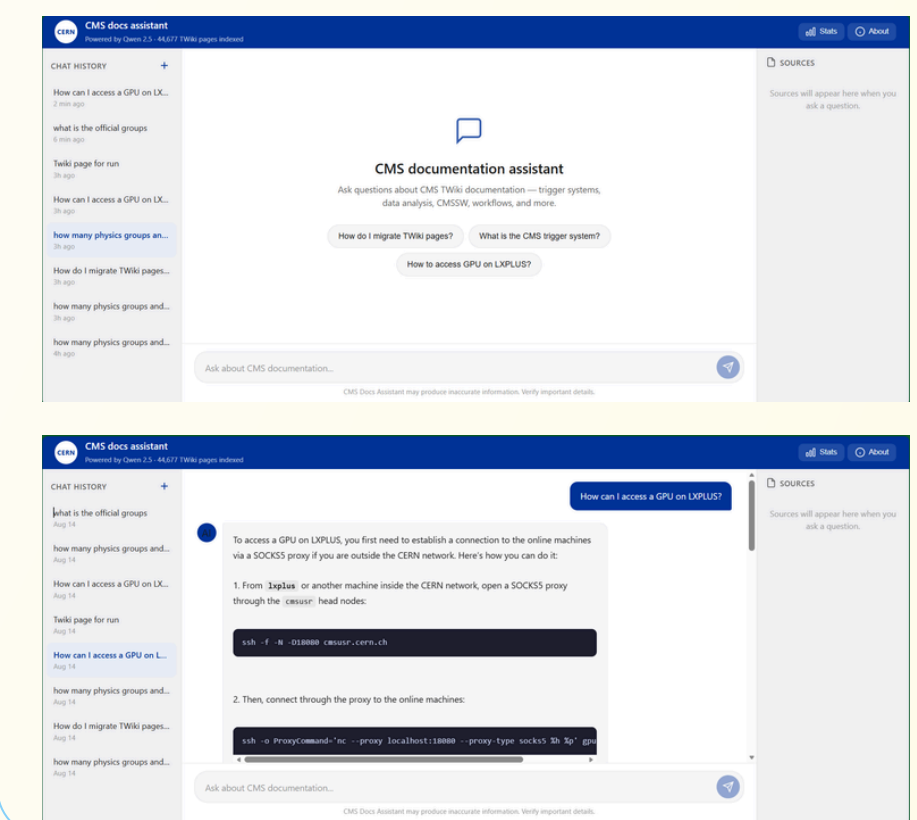
PART 2 : AI SEMANTIC SEARCH

Using RAG and LLM to build the answer

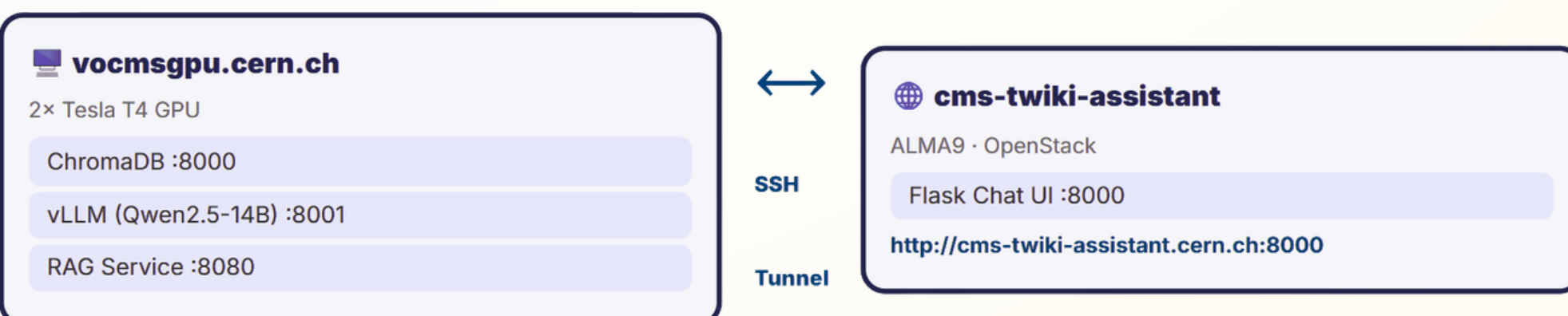


CHAT INTERFACE

Using RAG and LLM to build the answer



CERN CLOUD INFRASTRUCTURE



FUTURE WORK

- Strengthen the RAG Pipeline**
Explore GraphRAG, Agentic RAG, and further retrieval improvements.
- Harden the Deployment**
Introduce Docker, HTTPS, CERN SSO, and per-user sessions.
- Move Toward Production**
Integrate CERN's LiteLLM Gateway and establish long-term service ownership.



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